

ARThings – enhancing the visitors' experience in museums through collaborative AR

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ABSTRACT

Museum exhibitions are adopting Culturtainment, a new range of emergent media, as means of engaging their visitors through a high level of creative output and consumption, where Augmented Reality becomes a technological challenge for augmenting the visitors' experience in a meaningful way. ARThings defines its scope to extend the concept of touring in an art exhibition of paintings, towards collecting itinerary and interaction analytics, through collaborative AR. A demonstrative prototype has been implemented for the Art Museum of Braşov, Transylvania, Romania, where a User-Acceptance Test has been conducted on 21 participants, together with on-site assessment, positively validating a future adoption.

Keywords: museum exploration, visitor experience, AR scanning, fine art, social interaction

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Mixed/augmented reality; Human-centered computing—Collaborative and social computing—Collaborative and social computing systems and tools—Social tagging systems; Applied computing—Arts and humanities—Fine arts

1 INTRODUCTION

The use of new technologies demands a new type of literacy on behalf of the visitor [1], where two distinct new forms of museum experience are emerging [2]: 1. the narrative-driven, media-enhanced on-site experience - characterized by a multi-sensory, contextualized and immersive experience, bringing a diversity of people together and providing a collective experience; 2. the personalized media-driven off-site experience - characterized by a personalized, on-demand, global experience, enabling information sharing between individuals. The use of Augmented Reality (AR) in museums and exhibitions are adopted as practices and paradigms for SMART MUSEUMS development, where 2 main paradigms arise. The paradigm of offering a personalized experience impacts both the user engagement, as well as collecting a meaningful feedback that can be further used as curated content. The second paradigm relies on augmenting the cultural exploration with real time interaction that provides the visitor with access to extended descriptive information, complementing to the one provided in the physical format.

In such a context, the aim of ARThings is to provide a personalized experience for visitors, enabling social feedback throughout their itineraries, within a fine art exhibition of paintings, towards improving both the visiting experience, as well as providing

meaningful feedback to the exhibition curators, through using widely-available technology (AR, mobile deployment). The background of ARThings is defined by applied technology to fine art exploration and the use of AR and social feedback [3] to characterize the user perception in cultural explorations [4]. An assessment of subjective user feedback has been proposed as an experimental ARThings User-Acceptance Test, and conducted among both potential visitors and museum representatives, validating a further in-house adoption.

2 ARTHINGS - AUGMENTED REALITY AND SOCIAL INTERACTION IN CULTURAL EXPLORATION

In the context in which the museums are challenged by changes and are required to identify new means of attracting and engaging the visitors through various technological approaches, the ARThings provides an application-based solution that enhances the interaction with exhibition items in order to improve the perception on the exhibition item (i.e. painting). The implementation framework can be translated and applied to different cultural spaces and fine art exhibitions or curated galleries.

The implementation relies on a .PHP client-server architecture developed on a Unity3D Engine with Vuforia AR integration, where the main functionalities include: AR scanning of exhibition paintings, with depiction of regions of interest (RoI); user feedback via Likert scale, communication through client-server architecture and comments prioritization based on users' reactions. The technology is widely available for users with a deployment for mobile devices. Future integration of evaluation metrics could be applied to generate personalized experiences. The happy flow is initiated by the user when AR scanning the painting, where RoIs are displayed on the corresponding displayed image target. Each RoI is actionable and provides specific information, for depicted details of the painting. The user will be able to provide feedback on the information received and discuss with other art enthusiasts, through a chat section displayed once the AR graphic overlay becomes active.



Figure 1 – ARThings functionalities: 1. RoI with descriptive text and additional visual details 2. Interactive chat with emotion-based reactions 3. Comments' ranking based on a likert-scale

User feedback is collected at 2 levels: 1. an average rating collected for each RoI, that describes the user perception of the descriptive content, as shown in Figure 2; 2. a rating at users' comments level, based on which a ranking of the comments is being displayed as displayed in Figure 3.

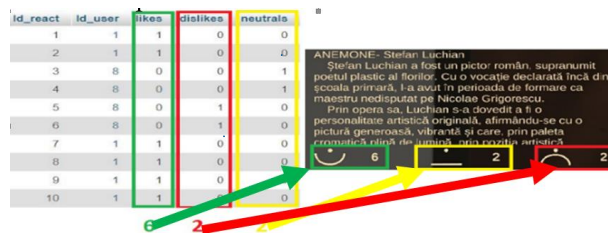


Figure 2 – User rating at RoI level for each painting

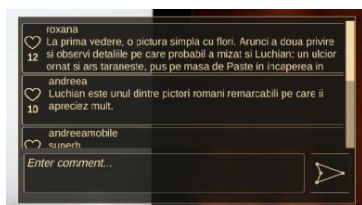


Figure 3 – Ranking of user comments for each painting

The prototype assessment has been conducted in the Art Museum of Braşov, providing a model of promoting local traditional artists and national heritage to both local and international visitors. The museum is located in Transylvania, Romania and houses an impressive collection of artworks, not only by the most famous and influential Romanian painters, such as Grigorescu and Tonita, but also by local undiscovered painters.

The 11 paintings that have been included in the UAT are part of the permanent collection of the museum and are located in one of the 3 main rooms of the museum, as shown in the Figure 4.

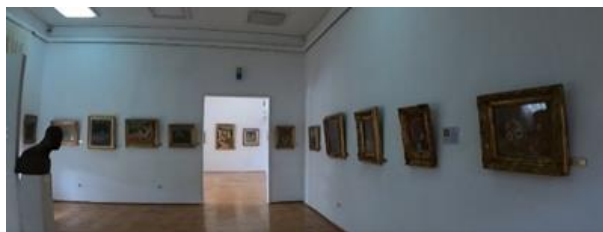


Figure 4 – The main hall and part of the permanent collection of the Art Museum of Braşov

The lighting conditions in the room did not impose issues on AR marker detection. However, given the wide open space and the interference between the artificial lightning with the exposure to natural daylight, such variations shall be taken into consideration in a future optimization to avoid inconsistent lighting between the real world stream and the overlaid virtual elements (i.e. painting RoIs, users' chat).

ARThings on-site validation has been performed under the approval of the museum's representatives. The experimental validation has been captured as a demonstrative walkthrough through the permanent collection, as exemplified in Figure 5.

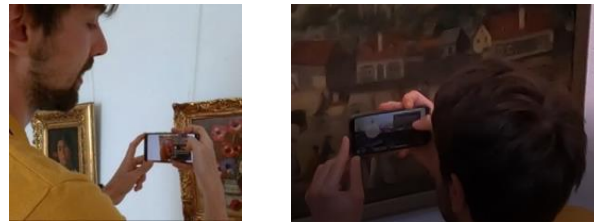


Figure 5 – Demonstrative museum walkthrough in different light conditions

Twenty-one respondents aged between 20 and 40 from Romania, with different artistic and cultural background, took part in the validation, testing the prototype and offering subjective open feedback. Results showed that the targeted market segment of tech-accustomed users aged 20-40 with artistic or cultural pursuit, would be interested in using this application with some modifications. One outcome of the assessment is the segmentation of the user profiles in 2 categories: 1. a profile of art enthusiasts who would like to know more about the exhibition items, as well as to receive informational recommendations; 2. a profile of a lower-level art consumers, engaged by tech-enabled art, who stated their preference of augmenting the app with gamification features and how such would increase their interest in spending more time in such a museum. In the Figure 12, we can see the division of participants of the study in numbers. 61,9% of the participants resulted to prefer a gamification experience, where 80% of respondents would choose for ARThings instead of an audio guide provided by the museum.

3 CONCLUSIONS

The positive evaluation of the ARThings prototype towards in-house adoption within the Museum of Braşov, demonstrates the need of including tech-based strategies into the marketing strategies of cultural and artistic spaces. Further development proposes the encapsulation of the AR module into a recommender system for personalized itineraries. With respect to the UAT, the current conducted study poses a limitation regarding the age interval included in the study (20-39). The segment of 40-55 years has been identified as being less represented in Romania, enhancing the need of improving the tech-adoption.

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